

**FIFTH SEMESTER U.G. (CUCBCSS—UG) DEGREE [SPECIAL]
EXAMINATION, NOVEMBER 2020**

Mathematics

MAT 5D 19—MATHEMATICS FOR SOCIAL SCIENCES

Time : Two Hours

Maximum : 40 Marks

Part A

*All questions to be attended.
Each question carries 1 mark.*

1. What is an odd function ?
2. What is the derivative of $\sin x$?
3. What is the solution of $x^2 - 4 = 0$?
4. Define exponential function.
5. Evaluate $\lim_{x \rightarrow 1} x^2 - 1$.
6. What is a linear function ?

(6 × 1 = 6 marks)

Part B

All questions can be attended and overall ceiling.

*Answer any **five** questions.
Each question carries 2 marks.*

7. Solve $2x^2 - 3x + 1 = 0$.
8. State mean value theorem.
9. Find the slope of the curve $ax^2 + bx + c$.
10. What is point of inflection ?
11. If a particle moves along a curve with velocity given by the function $f(x) = 5x^3 + 3x^2 - x + 4$ find the acceleration at $x = 1$.
12. Form the partial differential equation of $z = ax + by + c$.
13. If the marginal revenue function is given by $Q^2 + 12Q - 21$. Find the total revenue function.

(5 × 2 = 10 marks)

Turn over

Part C

All questions can be attended and overall ceiling.

*Answer any **three** questions.*

Each question carries 4 marks.

14. Evaluate $\int e^{2x} dx$.
15. Find the second partial derivatives of $z = x^3 + y^3 + 3xy$.
16. Find the sum and product of roots of equation $2x^2 + x - 3 = 0$.
17. If $f(x) = \sin x$ and $g(x) = 3x + 1$ find f.g. and g.f.
18. A particle is moving along a curve $S = 12t - 3t^2$. Find the velocity and acceleration at $t = 0$.
(3 × 4 = 12 marks)

Part D

All questions can be attended and overall ceiling.

*Answer any **two** questions.*

Each question carries 6 marks.

19. Differentiate x^n using first principle.
20. If $y = a \cos mx + b \sin nx$ show that $d^2y/dx^2 + n^2y = 0$.
21. Integrate the following : i) $\log x$; ii) $(4x^3 + 2x^2 + x - 3)/x^2$.
(2 × 6 = 12 marks)

FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021

(CBCSS—UG)

Mathematics

MTS 5B 05—THEORY OF EQUATIONS AND ABSTRACT ALGEBRA

(2019 Admissions)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A*Answer at least ten questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 30.*

1. Show that $x^5 - 3x^4 + x^2 - 2x - 3$ is divisible by $x - 3$.
2. Factorize into linear factors the polynomial $x^4 - 1$.
3. Write a cubic equation with roots $1, 1 + i, 1 - i$.
4. State Identity theorem.
5. How many real roots has the equation $x^4 - 4ax + b = 0$.
6. Make addition and multiplication tables for $\mathbb{Z}_2$.
7. Check whether the relation on $\mathbb{R}$ defined by $a \sim b$ if $a - b \in \mathbb{Q}$ is an equivalence relation.
8. Consider the permutations $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix}$ and $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$. Compute $\sigma \tau$ and $\tau \sigma$.
9. Let G be a group and $a, b \in G$. Show that $(ab)^{-1} = b^{-1}a^{-1}$.
10. Write a subgroup of $(\mathbb{Z}, +)$.

Turn over

11. Check whether $\mathbb{Z} \times \mathbb{Z}$ is cyclic.
12. Find order of the permutation $(1, 3)(2, 6)(1, 4, 5)$.
13. Let $\Phi : G_1 \rightarrow G_2$ be a group homomorphism. Show that $\Phi(e) = e'$ where e and e' are identity elements of G_1 and G_2 respectively.
14. Define a Ring.
15. Give example of an integral domain.

(10 × 3 = 30 marks)

Section B

Answer at least five questions.

Each question carries 6 marks.

All questions can be attended.

Overall Ceiling 30.

16. Solve $x^5 - 3x^4 + 4x^3 - 4x + 4$ having the root $1+i$.
17. Solve the cubic equation $2x^3 - x^2 - 18x + 9 = 0$ whose roots are a, b, c with $a + b = 0$.
18. Find an upper limit of the positive roots of the equation $2x^5 - 7x^4 - 5x^3 + 6x^2 + 3x - 10 = 0$.
19. Prove that set of all even permutations of S_n is a subgroup of S_n .
20. Define $*$ on $\mathbb{Z}$ by $a * b = a - b$. Check whether $(\mathbb{Z}, *)$ is a group.
21. Check whether $\mathbb{Z}_n$ is cyclic.
22. Draw the subgroup diagram of $\mathbb{Z}_{36}$.
23. Let G_1 and G_2 be groups and let $\Phi : G_1 \rightarrow G_2$ be a function such that $\Phi(ab) = \Phi(a)\Phi(b)$ for all $a, b \in G$. Prove that Φ is 1-1 if and only if $\Phi(x) = e$ implies that $x = e$ for all $x \in G_1$.

(5 × 6 = 30 marks)

Section C

*Answer any **two** questions.*

Each question carries 10 marks.

24. Examine whether $x^4 - x^3 - x^2 + 19x - 42 = 0$ has integral roots or not.

25. Solve $x^3 - 6x - 6 = 0$ by Cardan's method.

26. Let G be a cyclic group. Show that :

(a) If G is infinite then $G \cong \mathbb{Z}$.

(b) If $|G| = n$, then $G \cong \mathbb{Z}_n$.

27. State and prove Lagrange's theorem.

(2 × 10 = 20 marks)

FIFTH SEMESTER B.A./B.Sc. DEGREE EXAMINATION, NOVEMBER 2021**(CUCBCSS—UG)****Mathematics****MAT 5D 19—MATHEMATICS FOR SOCIAL SCIENCES****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 10****Maximum : 10 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 10.
2. The candidate should check that the question paper supplied to him/her contains all the 10 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 5D 19—MATHEMATICS FOR SOCIAL SCIENCES

(Multiple Choice Questions for SDE Candidates)

1. If $f(x) = x^2 + 2x + 6$ then $f(0)$ is _____.
- (A) 6. (B) 9.
(C) 0. (D) 5.
2. The limit of $f(x) = x^2 + x(x - 3)$ as $x \rightarrow 3$ is _____.
- (A) 3. (B) 9.
(C) 0. (D) -3.
3. If $f(x) = (2x + 5)^5$ then $f'(x)$ is _____.
- (A) $\frac{5}{(2x + 5)^5}$. (B) $5(2x + 5)^4$.
(C) $\frac{(2x + 5)^6}{12}$. (D) $10(2x + 5)^4$.
4. If $f(x) = 6x^3$ then $f''(x)$ is _____.
- (A) $\frac{5}{x^4}$. (B) $18x^2$.
(C) $36x$. (D) 36.
5. At $x = 2$ the function $f(x) = x + 5$ is _____.
- (A) Increasing. (B) Decreasing.
(C) Inflection. (D) Slope = 0.
6. The value of $\log_2 16$ is equivalent to _____.
- (A) 2 (B) 0.
(C) 4. (D) 8.

7. Derivative of $(x) = e^{2x^3}$ is _____.

(A) e^{6x^3} .

(B) $6x^2 e^{2x^3}$.

(C) e^{2x^3} .

(D) $\frac{e^{2x^3}}{6x}$.

8. The value of $\int 6e^{3x} dx$ is _____.

(A) $2e^{3x} + c$.

(B) $\frac{6}{e^{3x}} + c$.

(C) $18e^{3x} + c$.

(D) $\frac{6e^{4x}}{4} + c$.

9. If $z = e^{x^2y}$ then $\frac{\partial z}{\partial x}$ is _____.

(A) $x^2 e^{x^2y}$.

(B) $xy e^{x^2y}$.

(C) $2xy e^{x^2y}$.

(D) $\frac{x^2 y}{3} e^{x^2y}$.

10. Using L'Hospital's rule the limit of $f(x) = \frac{x^2 - 7x}{x^3 + 2x}$ as $x \rightarrow 0$ is _____.

(A) 0.

(B) 2.

(C) $\frac{-7}{2}$.

(D) 7.